Linear-time sorting exercises

08-E01 Comparison lower bound

Difficulty: Foundational.

Four distinct elements have 4! possible orders. Use a decision tree to give the minimum worst-case comparison lower bound. Does it exclude counting sort?

08-E02 Stable counting placement

Difficulty: Advanced.

For A=[(2,a),(0,b),(2,c),(1,d),(2,e)] with keys 0..2, compute frequencies, cumulative counts, and right-to-left placement. Verify stability.

08-E03 Radix cost tradeoff

Difficulty: Advanced.

For n nonnegative 32-bit integers, compare 8-bit and 16-bit digits using O(d(n+r)). Which additional resources constrain the choice?

08-E04 Exactly where stability fails

Difficulty: Advanced.

Trace [2a,0b,2c,1d,2e] when counting placement uses decrementing block endpoints but scans forward. Use [12,11] to explain stable digit sorting in LSD radix sort.

08-E05 Nonuniform bins and squared sizes

Difficulty: Research.

Four independent inputs choose four bins with probabilities [1/2,1/4,1/4,0]. Compute $E[\Sigma N_i^2]$ and compare uniform bins. Construct dependent records whose individual bin indices remain uniform.

08-E06 Prefix queries and negative-key shifts

Difficulty: Advanced.

For [−2,1,−2,0,3,1], build frequencies and prefixes for −2..3. Count records in closed ranges [−1,1] and [−2,−2]. Why is the placement-consumed count array unsuitable for these queries?

Work independently before opening the separate solutions PDF.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Linear-time sorting exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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